

Kazi Takia Talha

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SUMMARY

Senior-year Computer Science and Mathematics student graduating in 2027 with 16+ months of industry experience building machine learning systems, optimization models, and large-scale data pipelines.

EDUCATION

University of Toronto

Bachelor of Science (H.BSc), Double Major in Computer Science and Mathematics

Sep 2022 - Dec 2026

WORK EXPERIENCE

Hitachi Rail

Software Analyst Intern

May 2025 – Aug 2026

- Developed ETL and data processing pipelines for 2M+ ATS operational records, performing data cleaning, exploratory analysis, and feature engineering using SQL and statistical techniques (PCA, KneeLocator).
- Built and evaluated machine learning models for train travel time estimation, developing a benchmark from historical operational data to validate model performance and achieving an ~71% improvement over the existing approach.
- Designed a stochastic nonlinear optimization model for dwell time recommendation, reducing train energy consumption by 13% in performance testing while supporting delay recovery; developed function prototypes and software architecture specifications for integration into the existing codebase.
- Prototyped an AI-assisted data analysis workflow using MCP servers and LLMs, enabling users to interact with machine learning pipelines through natural language and automatically execute data processing and modeling tasks.
- Supported the Train Autonomy Platform, a sensor-based localization platform, by analyzing sensor data to validate radar ghost-target simulations, investigate detection-loss behavior, and prototype IMU slow-drift detection algorithms.

University of Toronto RLEA Lab

Undergraduate Research Volunteer

Nov 2025 – Present

- Evaluated instance segmentation models (Cellpose, SAM-based models) and analyzed segmentation quality to separate single-cell phytoliths from multi-cell microscopy images.
- Explored active learning pipelines to reduce labeling effort and improve segmentation performance on limited biological datasets.
- Investigated CNN- and Vision Transformer-based models for single-cell classification tasks.

The Hospital for Sick Children (Sick Kids)

Software Developer Intern

Jan 2025 – Apr 2025

- Built a full-stack web application for researchers in the Thrombosis Program to automate medical compression device wear-time detection, replacing manual sensor log review workflows with an interactive dashboard for data upload, visualization, and analysis of device usage patterns.
- Designed a statistical algorithm using multi-sensor time-series data (touch, EDA, heart rate, activity index) to predict device usage periods.

University of Toronto

Teaching Assistant

Sep 2024 – Present

- Held weekly office hours and tutorials; graded assignments and exams to ensure timely feedback and accurate assessment.

PROJECTS

NucleAIse

[View Project](#)

- Developed a baseline Python model using Pandas and scikit-learn to identify dataset issues and benchmark performance against ensemble models, achieving 73% accuracy.
- Designed a hierarchical SVM combined with Naive Bayes in Python to incorporate protein hierarchy into function prediction, improving classification precision.

Math-BERT-T5 Hybrid Model for Mathematical Reasoning

[View Project](#)

- Designed and implemented a six-layer encoder inspired by the T5 architecture for mathematical reasoning tasks.
- Analyzed 10+ research papers about language models to convert mathematical word problems into tokens and documented design trade-offs to reduce training complexity.

ADDITIONAL

Technical Skills: C++, Python, Java, C, JavaScript, PostgreSQL, Grafana, FastAPI, Docker, Linux, Git, Flask, JIRA

Data & ML: Pandas, NumPy, Power BI, MLflow, Data Visualization, PyTorch, TensorFlow, Scikit-learn, lakeFS